

CD4069UB CMOS hex inverter

1 Features

- Standardized symmetrical output characteristics
- Medium speed operation: t_{PHL} , t_{PLH} = 30 ns at 10 V (Typical)
- 100% Tested for quiescent current at 20 V
- Maximum input current of 1 μ A at 18 V over full package-temperature range, 100 nA at 18 V and 25°C
- Meets all requirements of JEDEC tentative standard No. 13B, *Standard Specifications for Description of B Series CMOS Devices*

2 Applications

- Logic inversion
- Pulse shaping
- Oscillators
- High-input-impedance amplifiers

3 Description

The CD4069UB device consist of six CMOS inverter circuits. These devices are intended for all general-purpose inverter applications where the medium-power TTL-drive and logic-level-conversion capabilities of circuits such as the CD4009 and CD4049 hex inverter and buffers are not required.

Device Information⁽¹⁾

PART NUMBER	PACKAGE (PINS)	BODY SIZE (NOM)
CD4069UBE	PDIP (14)	19.30 mm × 6.35 mm
CD4069UBF	CDIP (14)	19.56 mm × 6.67 mm
CD4069UBM	SOIC (14)	8.65 mm × 3.91 mm
CD4069UBNSR	SO (14)	10.30 mm × 5.30 mm
CD4069UBPW	TSSOP (14)	5.00 mm × 4.40 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

CD4069UB Functional Diagram

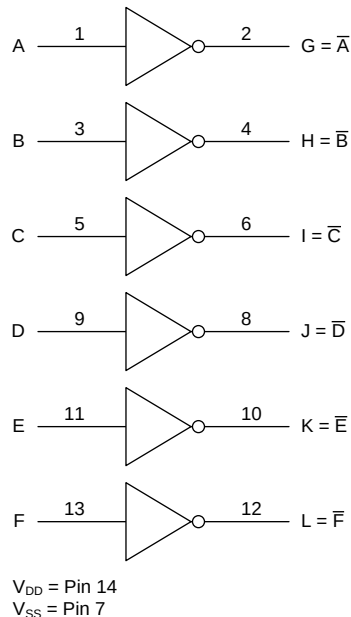


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4 Revision History

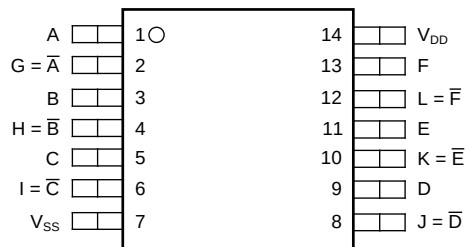
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision D (February 2016) to Revision E	Page
• Removed artifact "–" at t_{PHL} term on the second Features bullet	1
• Corrected V_I spec MIN/MAX values in the Abs Max Ratings table	4
• Corrected parameter I_{DD} max term to I_{DD} in the Elec Characteristics table	5
• Corrected parameter I_{OL} min term to I_{OL} in the Elec Characteristics table	5
• Corrected parameter V_{OL} max term to V_{OL} in the Elec Characteristics table	6
• Corrected parameter V_{IL} max term to V_{IL} in the Elec Characteristics table	6
• Corrected parameter V_{IH} min term to V_{IH} in the Elec Characteristics table	6
• Corrected parameter I_{IN} max term to I_{IN} in the Elec Characteristics table	7
• Added Y-axis label to Figure 1 image object	8
• Changed text string from "– t_{PHL} " to "of t_{PHL} " in the Feature Description paragraph.	13

Changes from Revision C (August 2003) to Revision D	Page
• Added <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section	1

5 Pin Configuration and Functions

D, J, N, NS, and PW Packages
14-Pin PDIP, CDIP, SOIC, SO, and TSSOP
Top View



Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
A	1	I	A input
B	3	I	B input
C	5	I	C input
D	9	I	D input
E	11	I	E input
F	13	I	F input
$G = \bar{A}$	2	O	G output
$H = \bar{B}$	4	O	H output
$I = \bar{C}$	6	O	I output
$J = \bar{D}$	8	O	J output
$K = \bar{E}$	10	O	K output
$L = \bar{F}$	12	O	L output
V_{DD}	14	—	Positive supply
V_{SS}	7	—	Negative supply

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

			MIN	MAX	UNIT
V _{DD}	DC supply-voltage (voltages referenced to V _{SS} terminal)		–0.5	20	V
V _I	Input voltage, all inputs		–0.5	V _{DD} + 0.5	V
I _{IK}	DC input current, any one input		–10	10	mA
P _D	Power dissipation per package	–55°C to 100°C		500	mW
		100°C to 125°C	12	200	
	Device dissipation per output transistor			100	mW
Lead temperature ⁽²⁾				265	°C
T _J	Junction temperature			150	°C
T _{stg}	Storage temperature		–65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) During soldering at distance 1/16 inch ± 1/32 inch (1.59 mm ± 0.79 mm) from case for 10 s maximum

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±500
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±200

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V _{DD}	Supply voltage	3	18	V
T _A	Operating temperature	–55	125	°C

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		CD4069UB					UNIT
		D (SOIC)	J (CDIP)	N (PDIP)	NS (SO)	PW (TSSOP)	
		14 PINS	14 PINS	14 PINS	14 PINS	14 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	94.9	—	57.9	91.2	122.1	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	56.4	28.5	45.5	48.8	50.8	°C/W
R _{θJB}	Junction-to-board thermal resistance	49.2	—	37.7	50	63.8	°C/W
ψ _{JT}	Junction-to-top characterization parameter	21.1	—	30.6	15	6.3	°C/W
ψ _{JB}	Junction-to-board characterization parameter	48.9	—	37.6	49.6	63.3	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	N/A	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Electrical Characteristics – Dynamic

 $T_A = 25^\circ\text{C}$; input $t_r, t_f = 20\text{ ns}$; $C_L = 50\text{ pF}$; $R_L = 200\text{ k}\Omega$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH}, t_{PHL} Propagation delay time	$V_{DD} (V) = 5$		55	110	ns
	$V_{DD} (V) = 10$		30	60	
	$V_{DD} (V) = 15$		25	50	
t_{THL}, t_{TLH} Transition time	$V_{DD} (V) = 5$		100	200	ns
	$V_{DD} (V) = 10$		50	100	
	$V_{DD} (V) = 15$		40	80	
C_{IN} Input capacitance	Any input		10	15	pF

6.6 Electrical Characteristics – Static

 $T_A = 25^\circ\text{C}$; input $t_r, t_f = 20\text{ ns}$; $C_L = 50\text{ pF}$; $R_L = 200\text{ k}\Omega$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_{DD} Quiescent device current	$V_{IN} = 0\text{V or } 5\text{ V}, V_{DD} = 5\text{ V}$	$T_A = -55^\circ\text{C}$			0.25	μA
		$T_A = -40^\circ\text{C}$			0.25	
		$T_A = 25^\circ\text{C}$		0.01	0.25	
		$T_A = 85^\circ\text{C}$			7.5	
		$T_A = 125^\circ\text{C}$			7.5	
	$V_{IN} = 0\text{ or } 10\text{ V}, V_{DD} = 10\text{ V}$	$T_A = -55^\circ\text{C}$			0.5	
		$T_A = -40^\circ\text{C}$			0.5	
		$T_A = 25^\circ\text{C}$		0.01	0.5	
		$T_A = 85^\circ\text{C}$			15	
		$T_A = 125^\circ\text{C}$			15	
	$V_{IN} = 0\text{ or } 15\text{ V}, V_{DD} = 15\text{ V}$	$T_A = -55^\circ\text{C}$			1	
		$T_A = -40^\circ\text{C}$			1	
		$T_A = 25^\circ\text{C}$		0.01	1	
		$T_A = 85^\circ\text{C}$			30	
		$T_A = 125^\circ\text{C}$			30	
	$V_{IN} = 0\text{ or } 20\text{ V}, V_{DD} = 20\text{ V}$	$T_A = -55^\circ\text{C}$			5	
		$T_A = -40^\circ\text{C}$			5	
		$T_A = 25^\circ\text{C}$		0.02	5	
		$T_A = 85^\circ\text{C}$			150	
		$T_A = 125^\circ\text{C}$			150	
I_{OL} Output low (sink) current	$V_O = 0.4\text{ V}, V_{IN} = 5\text{ V}, V_{DD} = 5\text{ V}$	$T_A = -55^\circ\text{C}$	0.64			mA
		$T_A = -40^\circ\text{C}$	0.61			
		$T_A = 25^\circ\text{C}$	0.51	1		
		$T_A = 85^\circ\text{C}$	0.42			
		$T_A = 125^\circ\text{C}$	0.36			
	$V_O = 0.5\text{ V}, V_{IN} = 10\text{ V}, V_{DD} = 10\text{ V}$	$T_A = -55^\circ\text{C}$	1.6			
		$T_A = -40^\circ\text{C}$	1.5			
		$T_A = 25^\circ\text{C}$	1.3	2.6		
		$T_A = 85^\circ\text{C}$	1.1			
		$T_A = 125^\circ\text{C}$	0.9			
	$V_O = 1.5\text{ V}, V_{IN} = 15\text{ V}, V_{DD} = 15\text{ V}$	$T_A = -55^\circ\text{C}$	4.2			
		$T_A = -40^\circ\text{C}$	4			
		$T_A = 25^\circ\text{C}$	3.4	6.8		
		$T_A = 85^\circ\text{C}$	2.8			
		$T_A = 125^\circ\text{C}$	2.4			

Electrical Characteristics – Static (continued)

 $T_A = 25^\circ\text{C}$; input $t_r, t_f = 20\text{ ns}$; $C_L = 50\text{ pF}$; $R_L = 200\text{ k}\Omega$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_{OH} Output high (source) current	$V_O = 4.6\text{ V}, V_{IN} = 0\text{ V}, V_{DD} = 5\text{ V}$	$T_A = -55^\circ\text{C}$	-0.64			mA
		$T_A = -40^\circ\text{C}$	-0.61			
		$T_A = 25^\circ\text{C}$	-0.51	-1		
		$T_A = 85^\circ\text{C}$	-0.42			
		$T_A = 125^\circ\text{C}$	-0.36			
	$V_O = 2.5\text{ V}, V_{IN} = 0\text{ V}, V_{DD} = 5\text{ V}$	$T_A = -55^\circ\text{C}$	-2			
		$T_A = -40^\circ\text{C}$	-1.8			
		$T_A = 25^\circ\text{C}$	-1.6	-3.2		
		$T_A = 85^\circ\text{C}$	-1.3			
		$T_A = 125^\circ\text{C}$	-1.15			
	$V_O = 9.5\text{ V}, V_{IN} = 0\text{ V}, V_{DD} = 10\text{ V}$	$T_A = -55^\circ\text{C}$	-1.6			
		$T_A = -40^\circ\text{C}$	-1.5			
		$T_A = 25^\circ\text{C}$	-1.3	-2.6		
		$T_A = 85^\circ\text{C}$	-1.1			
		$T_A = 125^\circ\text{C}$	-0.9			
	$V_O = 13.5\text{ V}, V_{IN} = 0\text{ V}, V_{DD} = 15\text{ V}$	$T_A = -55^\circ\text{C}$	-4.2			
		$T_A = -40^\circ\text{C}$	-4			
		$T_A = 25^\circ\text{C}$	-3.4	-6.8		
		$T_A = 85^\circ\text{C}$	-2.8			
		$T_A = 125^\circ\text{C}$	-2.4			
V_{OL} Low-level output voltage	$V_{IN} = 5\text{ V}, V_{DD} = 5\text{ V}$	$T_A = 25^\circ\text{C}$		0	0.05	V
		All other temperatures			0.05	
	$V_{IN} = 10\text{ V}, V_{DD} = 10\text{ V}$	$T_A = 25^\circ\text{C}$		0	0.05	
		All other temperatures			0.05	
	$V_{IN} = 15\text{ V}, V_{DD} = 15\text{ V}$	$T_A = 25^\circ\text{C}$		0	0.05	
		All other temperatures			0.05	
V_{OH} High-level output voltage	$V_{IN} = 0\text{ V}, V_{DD} = 5\text{ V}$	$T_A = 25^\circ\text{C}$	4.95	5		V
		All other temperatures	4.95			
	$V_{IN} = 0\text{ V}, V_{DD} = 10\text{ V}$	$T_A = 25^\circ\text{C}$	9.95	10		
		All other temperatures	9.95			
	$V_{IN} = 0\text{ V}, V_{DD} = 15\text{ V}$	$T_A = 25^\circ\text{C}$	14.95	15		
		All other temperatures	14.95			
V_{IL} Input low voltage	$V_O = 4.5\text{ V}, V_{DD} = 5\text{ V}, \text{all temperatures}$				1	V
	$V_O = 9\text{ V}, V_{DD} = 10\text{ V}, \text{all temperatures}$				2	
	$V_O = 13.5\text{ V}, V_{DD} = 15\text{ V}, \text{all temperatures}$				2.5	
V_{IH} Input high voltage	$V_O = 0.5\text{ V}, V_{DD} = 5\text{ V}, \text{all temperatures}$		4			V
	$V_O = 1\text{ V}, V_{DD} = 10\text{ V}, \text{all temperatures}$		8			
	$V_O = 1.5\text{ V}, V_{DD} = 15\text{ V}, \text{all temperatures}$		12.5			

Electrical Characteristics – Static (continued)

$T_A = 25^\circ\text{C}$; input $t_r, t_f = 20\text{ ns}$; $C_L = 50\text{ pF}$; $R_L = 200\text{ k}\Omega$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_{IN} Input current	$V_{IN} = 0\text{ V to }18\text{ V}, V_{DD} = 18\text{ V}$	$T_A = -55^\circ\text{C}$			± 01	μA
		$T_A = -40^\circ\text{C}$			± 01	
		$T_A = 25^\circ\text{C}$		$\pm 10^{-5}$	± 1	
		$T_A = 85^\circ\text{C}$			± 1	
		$T_A = 125^\circ\text{C}$			± 1	

6.7 Typical Characteristics

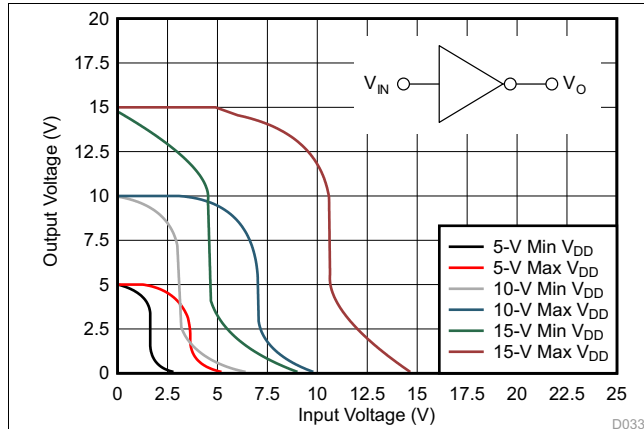


Figure 1. Minimum and Maximum Voltage Transfer Characteristics

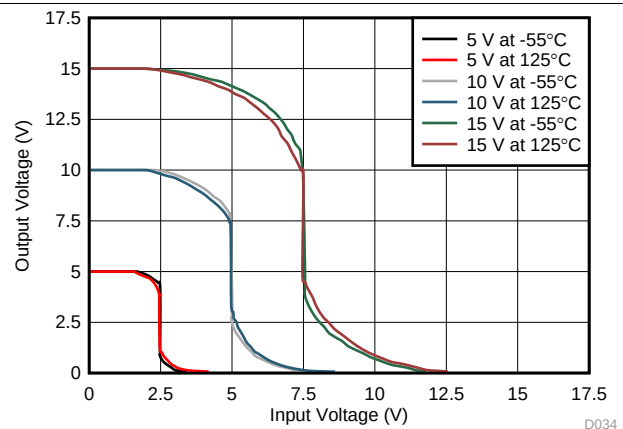


Figure 2. Typical Voltage Transfer Characteristics as a Function of Temperature

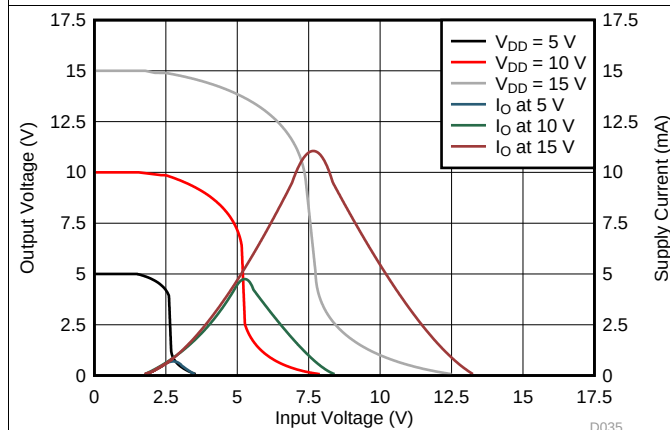


Figure 3. Typical Current and Voltage Transfer Characteristics

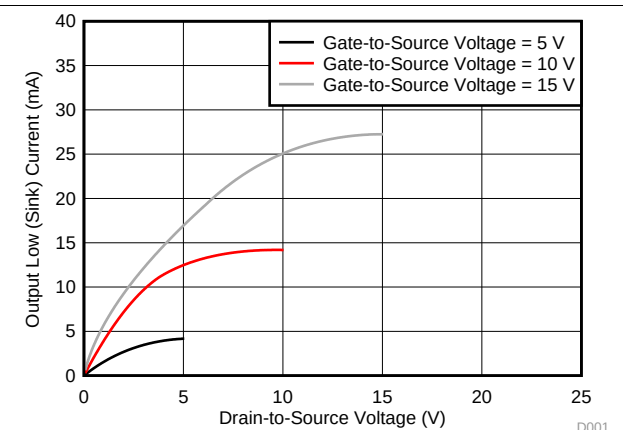


Figure 4. Typical Output Low (Sink) Current Characteristics

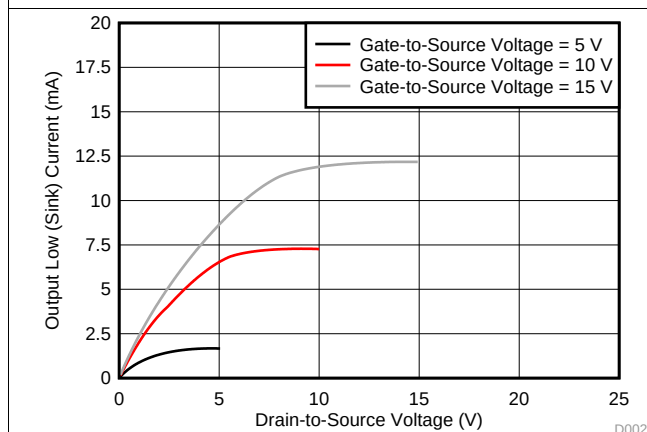


Figure 5. Minimum Output Low (Sink) Current Characteristics

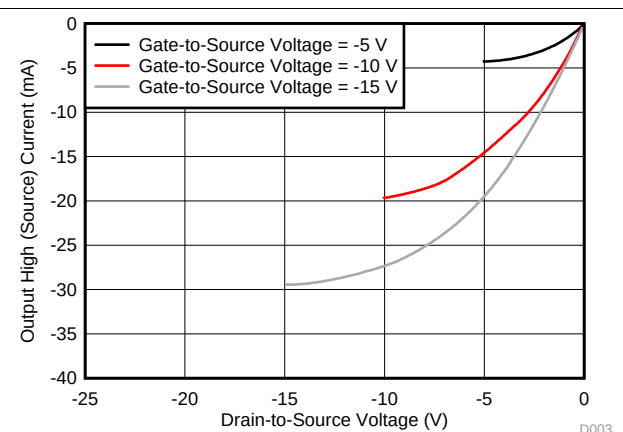


Figure 6. Typical Output High (Source) Current Characteristics

Typical Characteristics (continued)

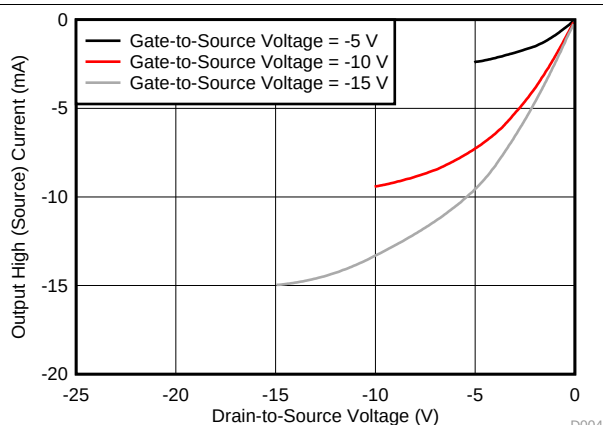


Figure 7. Minimum Output High (Source) Current Characteristics

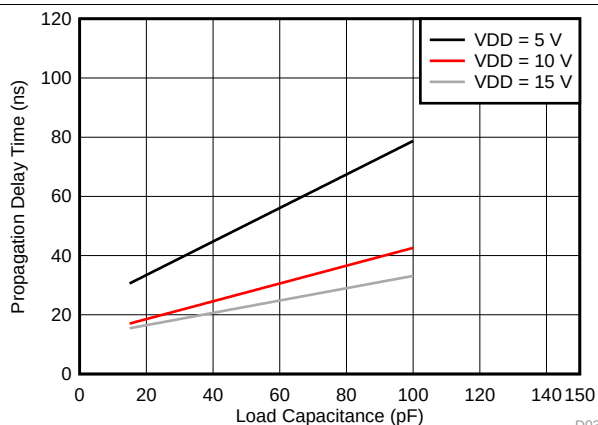


Figure 8. Typical Propagation Delay Time vs Load Capacitance

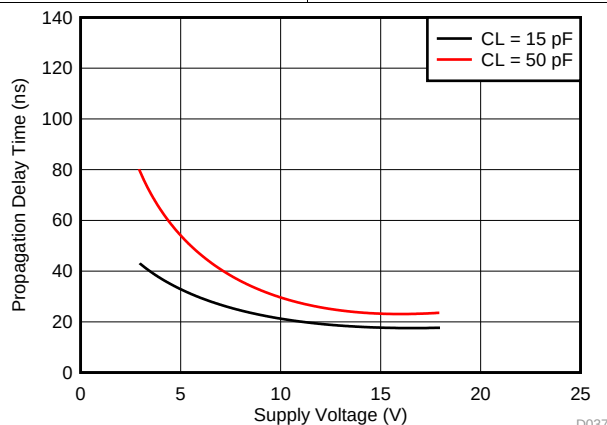


Figure 9. Typical Propagation Delay Time vs Supply Voltage

7 Parameter Measurement Information

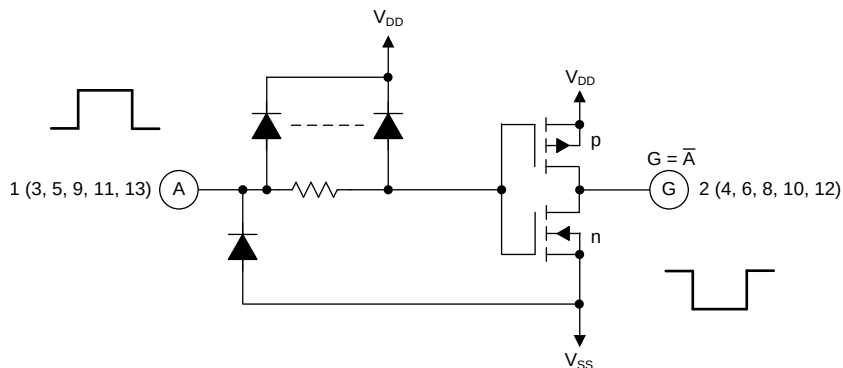


Figure 10. Schematic Diagram of One of Six Identical Inverters

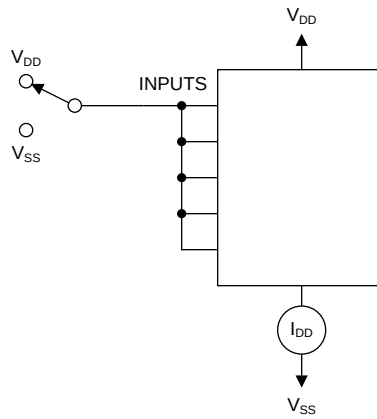


Figure 11. Quiescent Device Current Test Circuit

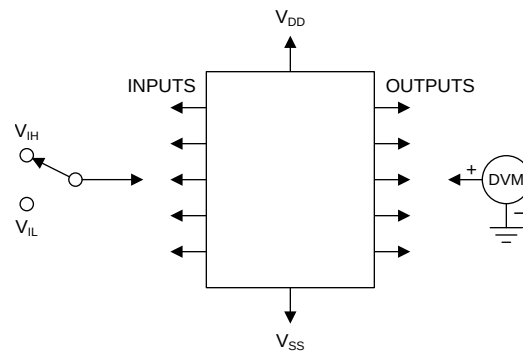


Figure 12. Noise Immunity Test Circuit

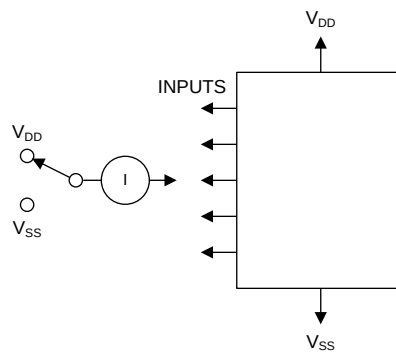


Figure 13. Input Leakage Current Test Circuit

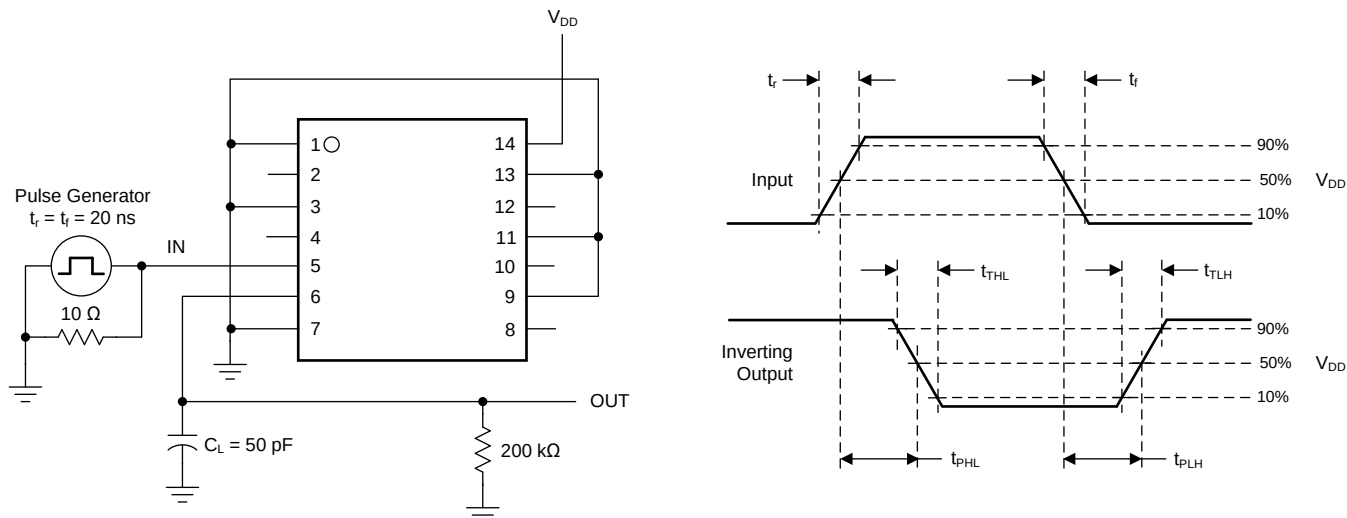


Figure 14. Dynamic Electrical Characteristics Test Circuit and Waveform

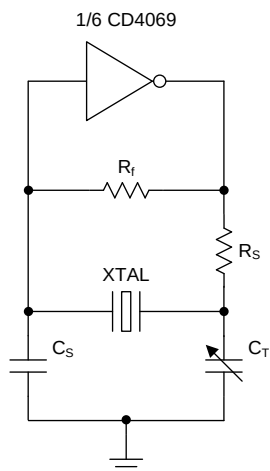


Figure 15. Typical Crystal Oscillator Circuit

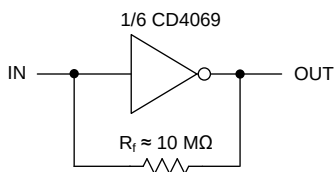


Figure 16. High-Input Impedance Amplifier

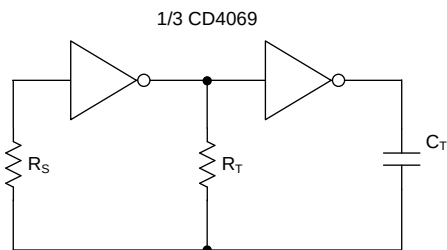
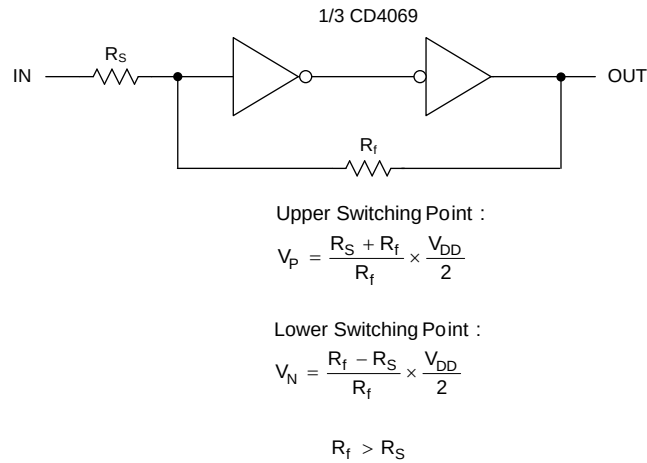
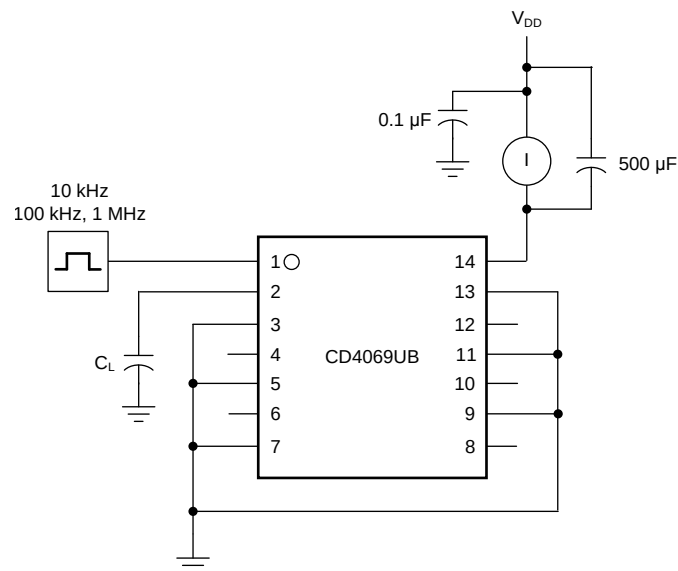


Figure 17. Typical RC Oscillator Circuit

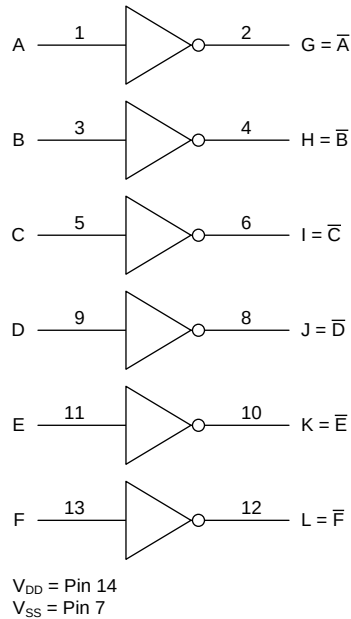

Figure 18. Input Pulse Shaping Circuit

Figure 19. Dynamic Power Dissipation Test Circuit

8 Detailed Description

8.1 Overview

The CD4069UB device has six inverter circuits. The recommended operating range is from 3 V to 18 V. The CD4069UB-series types are supplied in 14-pin hermetic dual-in-line ceramic packages (F3A suffix), 14-pin dual-in-line plastic packages (E suffix), 14-pin small-outline packages (M, MT, M96, and NSR suffixes), and 14-pin thin shrink small-outline packages (PW and PWR suffixes).

8.2 Functional Block Diagram



8.3 Feature Description

CD4069UB has standardized symmetrical output characteristics and a wide operating voltage range from 3 V to 18 V with quiescent current tested at 20 V. This has a medium operation speed of t_{PHL} , $t_{PLH} = 30$ ns (typical) at 10 V. The operating temperature is from -55°C to 125°C . CD4069UB meets all requirements of JEDEC tentative standard No. 13B, *Standard Specifications for Description of B Series CMOS Devices*.

8.4 Device Functional Modes

Table 1 shows the functional modes for CD4069UB.

Table 1. Function Table

INPUT A, B, C, D, E, F	OUTPUT G, H, I, J, K, L
H	L
L	H

9 Application and Implementation

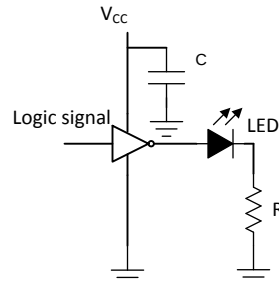
NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The CD4069UB device has a low input current of 1 μA at 18 V over full package-temperature range and 100 nA at 18 V, 25°C. This device has a wide operating voltage range from 3 V to 18 V and used in high voltage applications.

9.2 Typical Application



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Figure 20. CD4069UB Application

9.2.1 Design Requirements

The CD4069UB device is the industry's highest logic inverter operating at 18 V under recommended conditions. The lower drive capabilities makes it suitable for driving light loads like LED and greatly reduces chances of overshoots and undershoots.

9.2.2 Detailed Design Procedure

The recommended input conditions for [Figure 20](#) includes rise time and fall time specifications (see $\Delta t/\Delta V$ in [Recommended Operating Conditions](#)) and specified high and low levels (see V_{IH} and V_{IL} in [Recommended Operating Conditions](#)). Inputs are not overvoltage tolerant and must be below V_{CC} level because of the presence of input clamp diodes to V_{CC} .

The recommended output condition for the CD4069UB application includes specific load currents. Load currents must be limited so as to not exceed the total power (continuous current through V_{CC} or GND) for the device. These limits are located in the [Absolute Maximum Ratings](#). Outputs must not be pulled above V_{CC} .

Typical Application (continued)

9.2.3 Application Curves

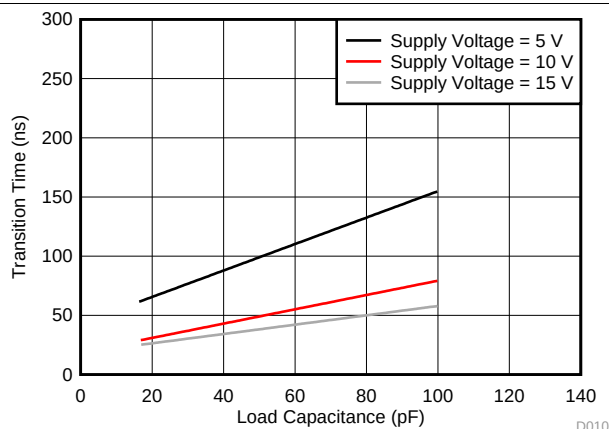


Figure 21. Typical Transition Time vs Load Capacitance

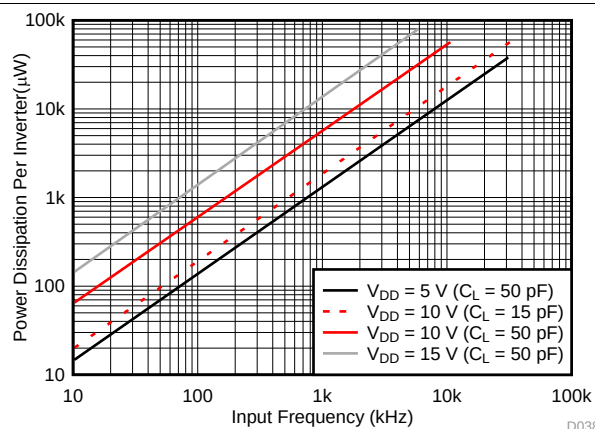


Figure 22. Typical Dynamic Power Dissipation vs Frequency

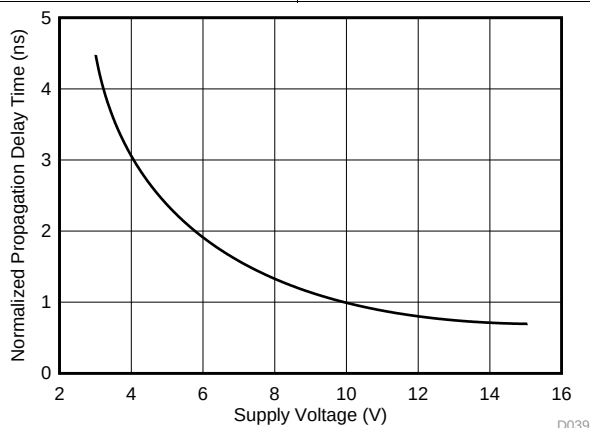


Figure 23. Variation of Normalized Propagation Delay Time (t_{PHL} and t_{PLH}) With Supply Voltage

10 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in [Recommended Operating Conditions](#).

Each V_{CC} pin must have a good bypass capacitor to prevent power disturbance. For devices with a single supply, TI recommends a 0.1- μ F capacitor. If there are multiple V_{CC} pins, then TI recommends a 0.01- μ F or 0.022- μ F capacitor for each power pin. It is acceptable to parallel multiple bypass capacitors to reject different frequencies of noise. 0.1- μ F and 1- μ F capacitors are commonly used in parallel. The bypass capacitor must be installed as close to the power pin as possible for best results.

11 Layout

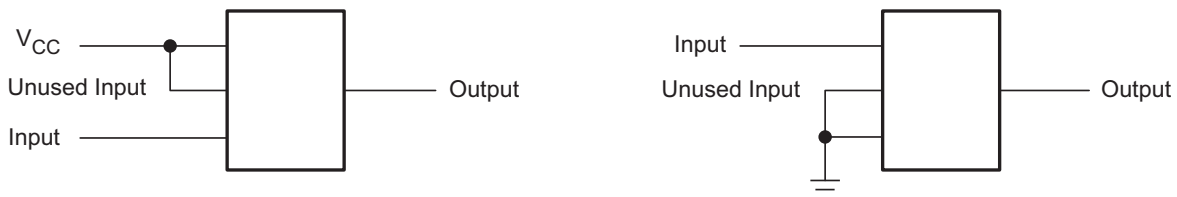
11.1 Layout Guidelines

When using multiple bit logic devices, inputs must never float.

In many cases, digital logic device functions or parts of these functions are unused (for example, when only two inputs of a triple-input and gate are used, or only 3 of the 4 buffer gates are used). Such input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. This rule must be observed under all circumstances specified in the next paragraph.

All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. See the application note, *Implications of Slow or Floating CMOS Inputs* ([SCBA004](#)), for more information on the effects of floating inputs. The logic level must apply to any particular unused input depending on the function of the device. Generally, they are tied to GND or V_{CC} (whichever is convenient).

11.2 Layout Example



12 Device and Documentation Support

12.1 Device Support

12.1.1 Third-Party Products Disclaimer

TI'S PUBLICATION OF INFORMATION REGARDING THIRD-PARTY PRODUCTS OR SERVICES DOES NOT CONSTITUTE AN ENDORSEMENT REGARDING THE SUITABILITY OF SUCH PRODUCTS OR SERVICES OR A WARRANTY, REPRESENTATION OR ENDORSEMENT OF SUCH PRODUCTS OR SERVICES, EITHER ALONE OR IN COMBINATION WITH ANY TI PRODUCT OR SERVICE.

12.2 Documentation Support

12.2.1 Related Documentation

For related documentation see the following:

- *Implications of Slow or Floating CMOS Inputs*, [SCBA004](#)

12.3 Community Resource

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

12.4 Trademarks

E2E is a trademark of Texas Instruments.
All other trademarks are the property of their respective owners.

12.5 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.6 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CD4069UBE	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD4069UBE
CD4069UBE.A	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD4069UBE
CD4069UBEE4	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD4069UBE
CD4069UBF	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	CD4069UBF
CD4069UBF.A	Active	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	CD4069UBF
CD4069UBF3A	NRND	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	CD4069UBF3A
CD4069UBF3A.A	NRND	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	CD4069UBF3A
CD4069UBM	Active	Production	SOIC (D) 14	50 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UBM
CD4069UBM.A	Active	Production	SOIC (D) 14	50 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UBM
CD4069UBM96	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UBM
CD4069UBM96.A	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UBM
CD4069UBMT	Obsolete	Production	SOIC (D) 14	-	-	Call TI	Call TI	-55 to 125	CD4069UBM
CD4069UBNSR	Active	Production	SOP (NS) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UB
CD4069UBNSR.A	Active	Production	SOP (NS) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4069UB
CD4069UBPW	Obsolete	Production	TSSOP (PW) 14	-	-	Call TI	Call TI	-55 to 125	CM069UB
CD4069UBPWR	Active	Production	TSSOP (PW) 14	2000 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-55 to 125	CM069UB
CD4069UBPWR.A	Active	Production	TSSOP (PW) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM069UB
CD4069UBPWRG4.A	Active	Production	TSSOP (PW) 14	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM069UB
JM38510/17401BCA	NRND	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	JM38510/ 17401BCA
JM38510/17401BCA.A	NRND	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	JM38510/ 17401BCA
M38510/17401BCA	NRND	Production	CDIP (J) 14	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	JM38510/ 17401BCA

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

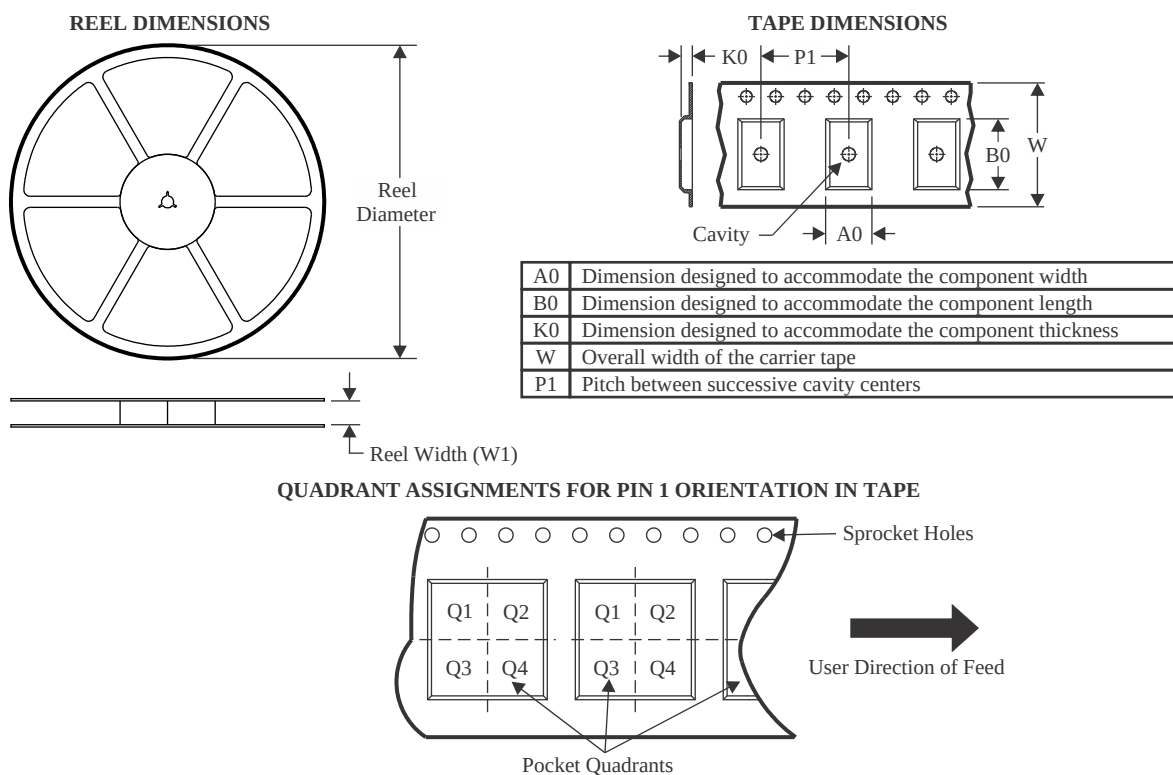
OTHER QUALIFIED VERSIONS OF CD4069UB, CD4069UB-MIL :

- Catalog : [CD4069UB](#)
- Military : [CD4069UB-MIL](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

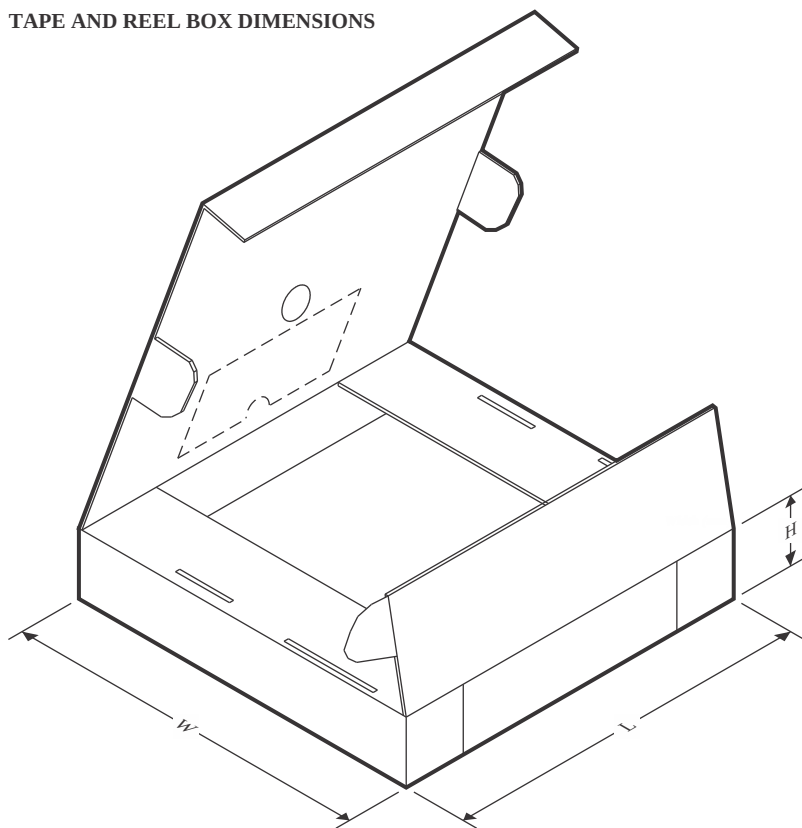
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CD4069UBM96	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CD4069UBNSR	SOP	NS	14	2000	330.0	16.4	8.1	10.4	2.5	12.0	16.0	Q1
CD4069UBPWR	TSSOP	PW	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
CD4069UBPWR	TSSOP	PW	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

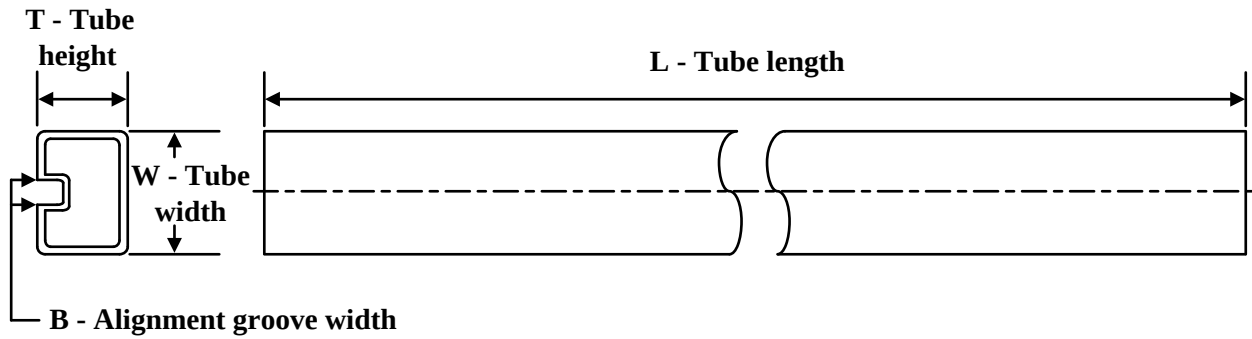
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

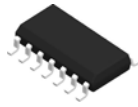
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CD4069UBM96	SOIC	D	14	2500	353.0	353.0	32.0
CD4069UBNSR	SOP	NS	14	2000	353.0	353.0	32.0
CD4069UBPWR	TSSOP	PW	14	2000	353.0	353.0	32.0
CD4069UBPWR	TSSOP	PW	14	2000	356.0	356.0	35.0

TUBE

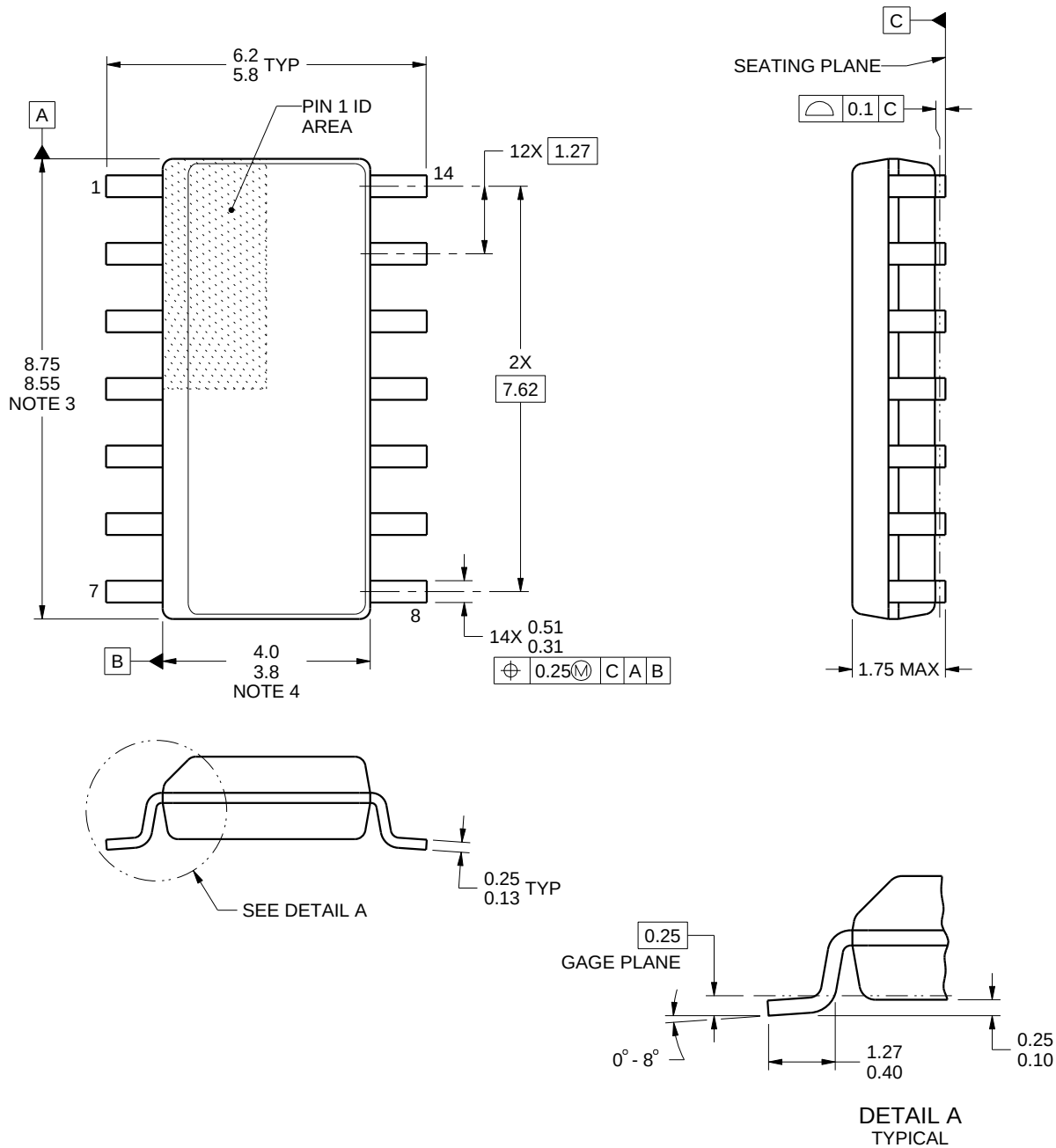


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
CD4069UBE	N	PDIP	14	25	506	13.97	11230	4.32
CD4069UBE.A	N	PDIP	14	25	506	13.97	11230	4.32
CD4069UBEE4	N	PDIP	14	25	506	13.97	11230	4.32
CD4069UBM	D	SOIC	14	50	506.6	8	3940	4.32
CD4069UBM.A	D	SOIC	14	50	506.6	8	3940	4.32

D0014A**PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



4220718/A 09/2016

NOTES:

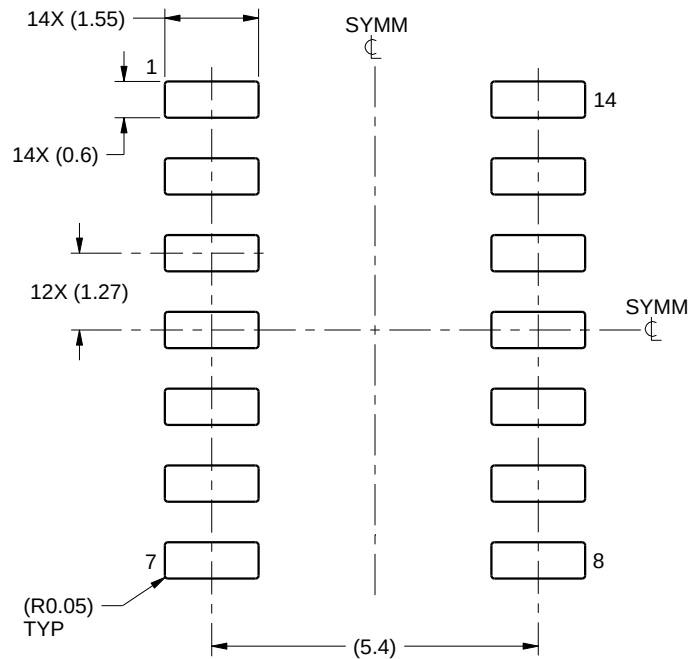
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

EXAMPLE BOARD LAYOUT

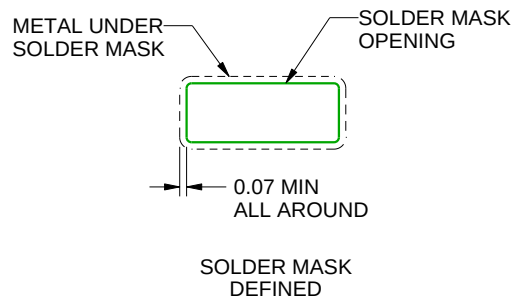
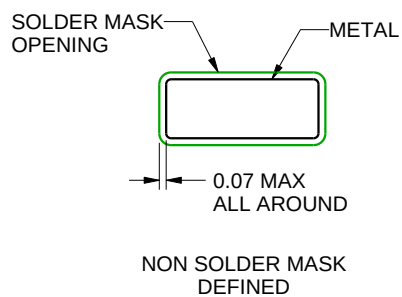
D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

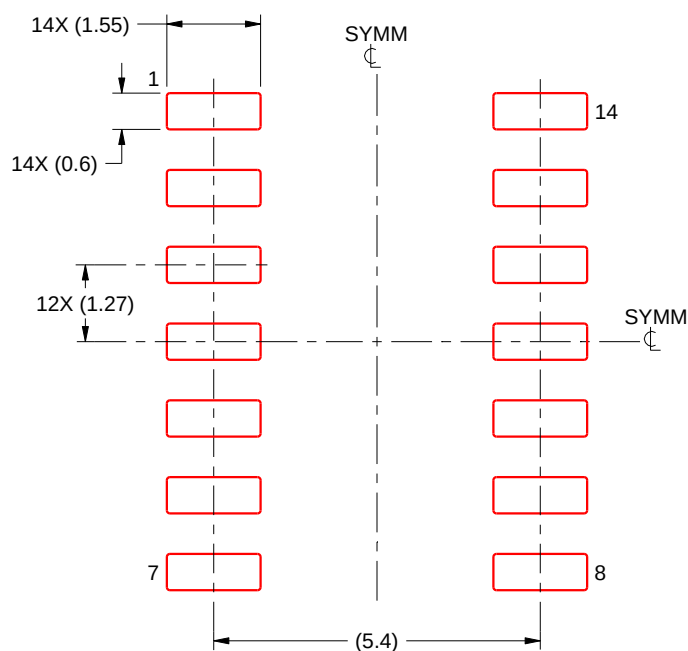
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

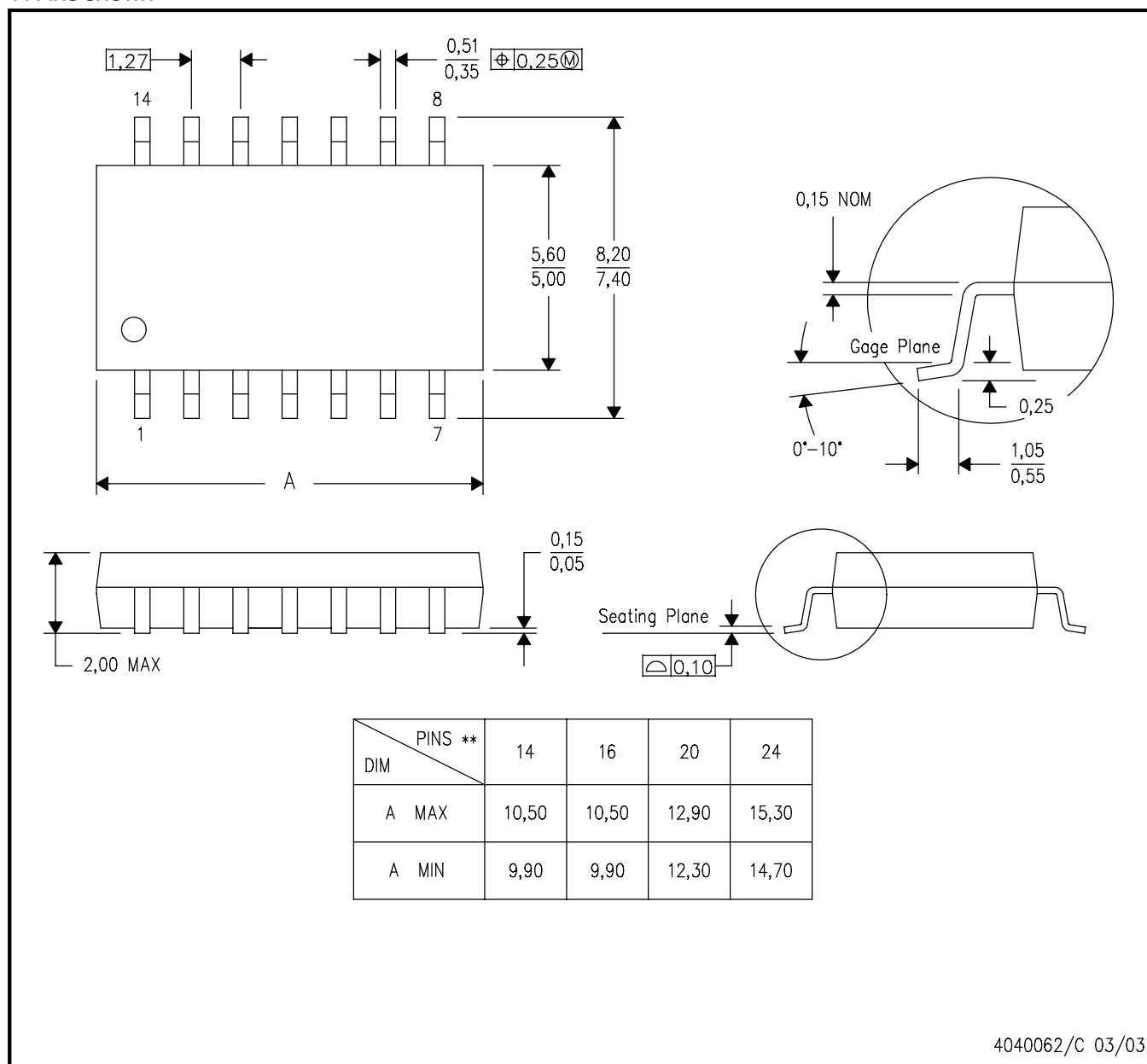
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

MECHANICAL DATA

NS (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN



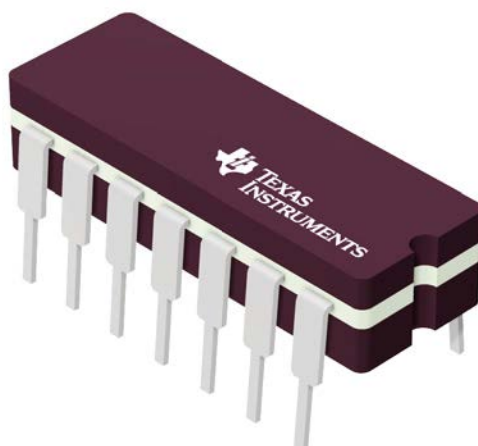
- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

J 14

GENERIC PACKAGE VIEW

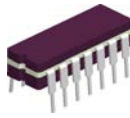
CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE

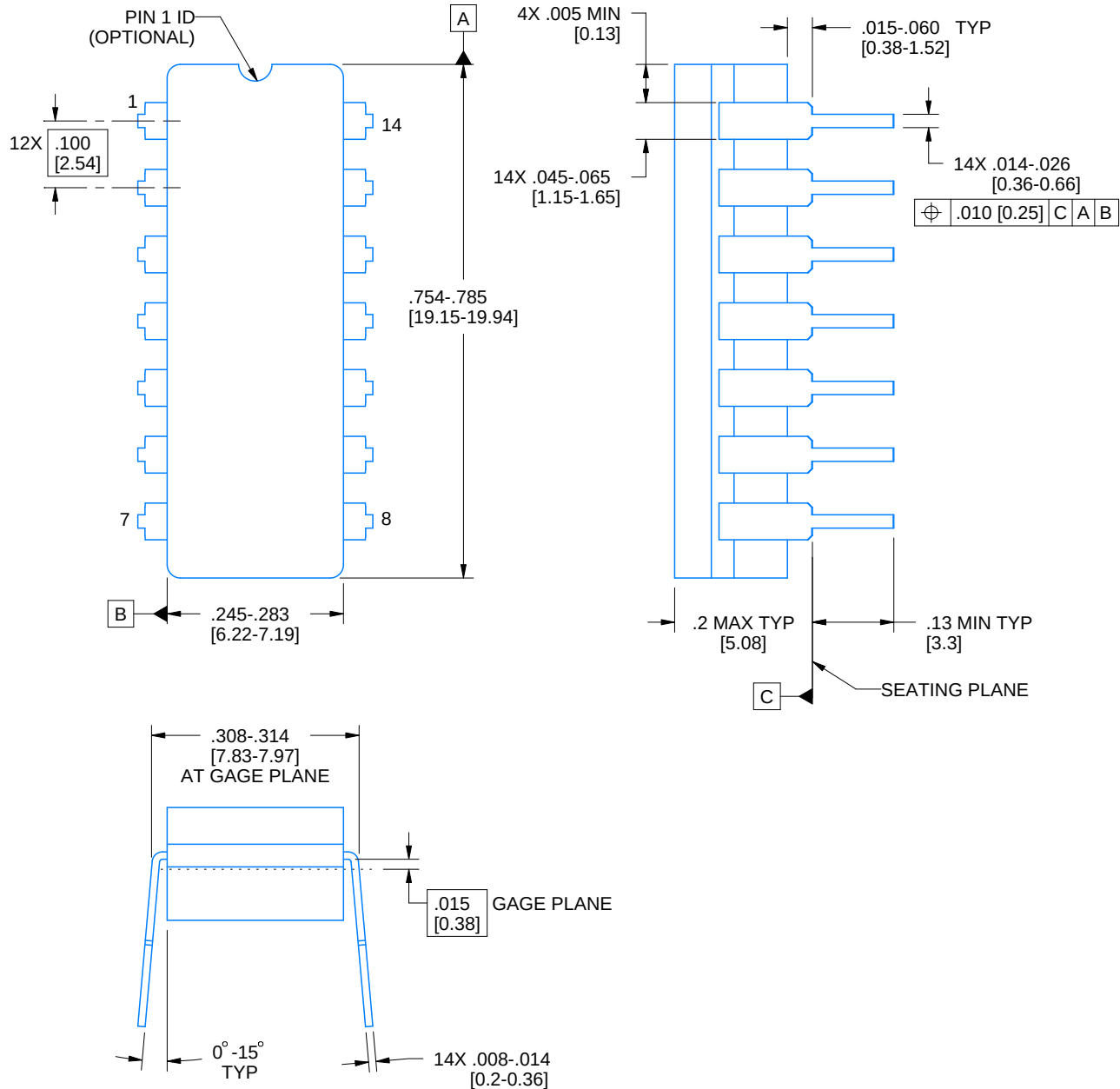


Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4040083-5/G

J0014A**PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

NOTES:

1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.

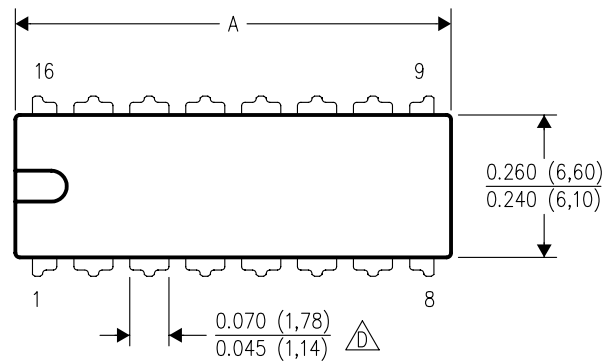


**TEXAS
INSTRUMENTS**
www.ti.com

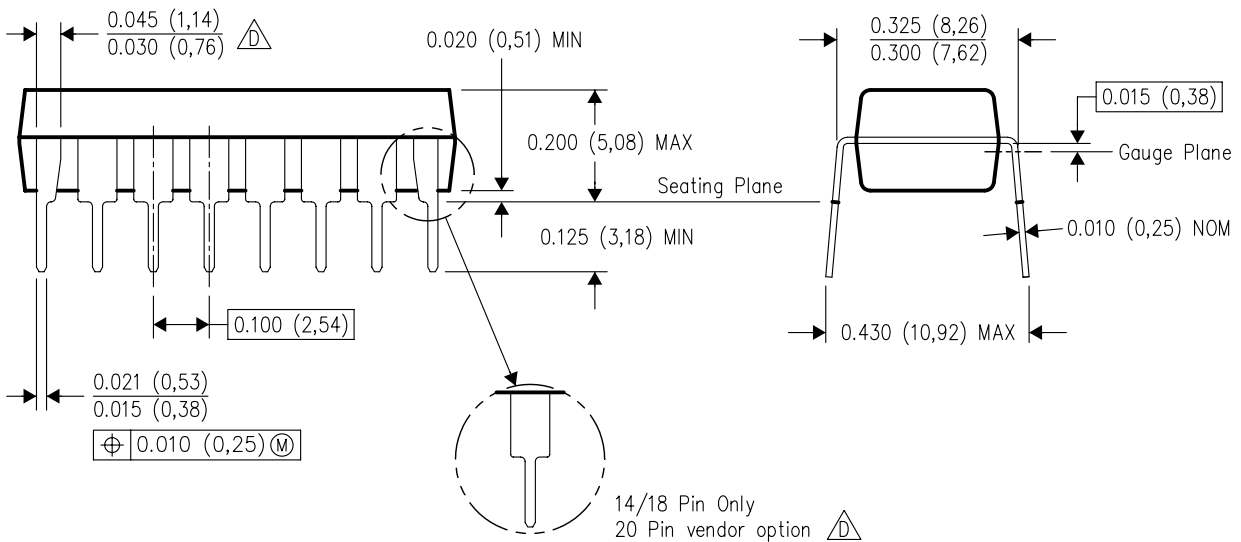
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



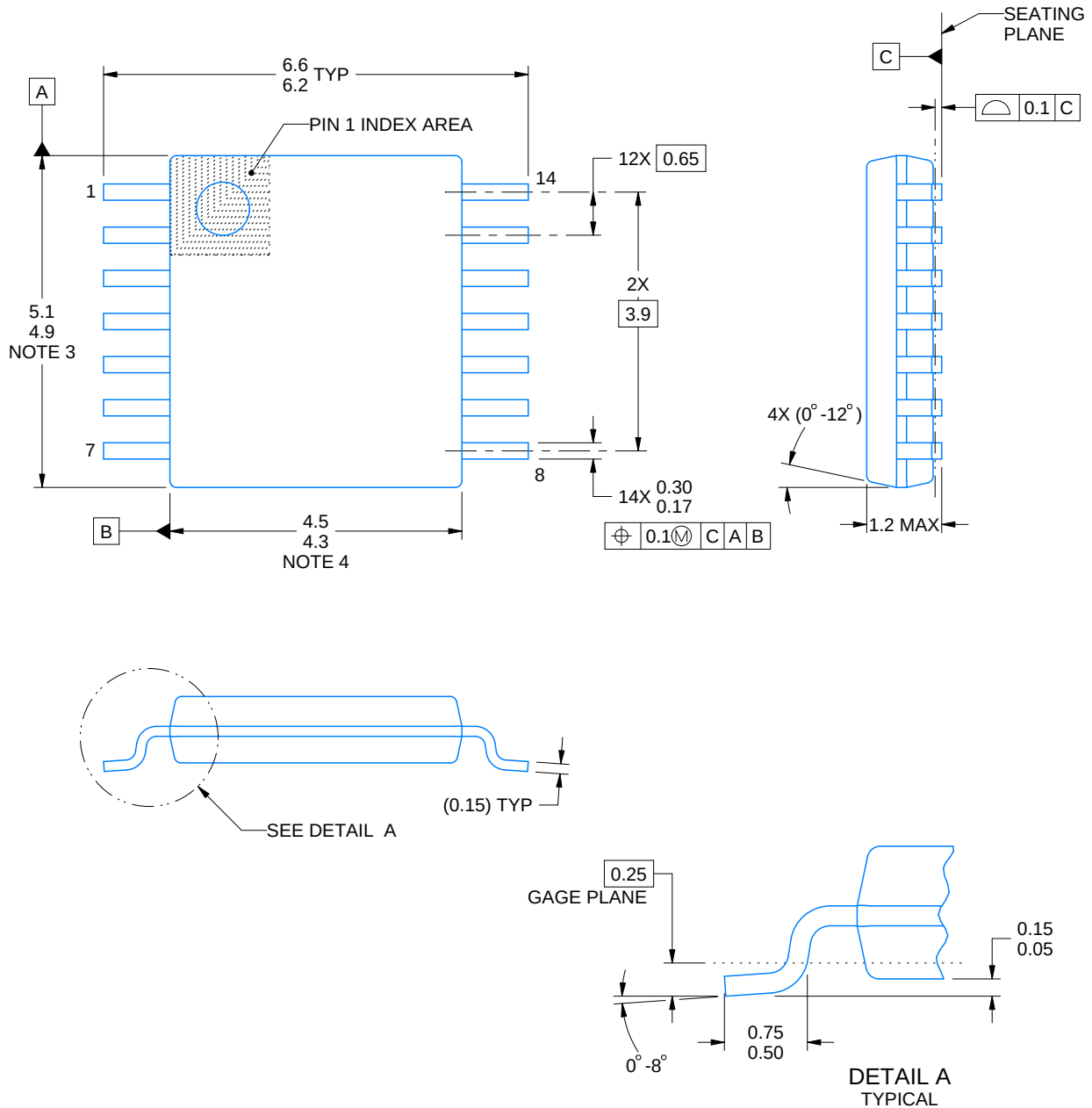
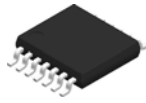
PINS **	14	16	18	20
DIM				
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



14/18 Pin Only
20 Pin vendor option

4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 - The 20 pin end lead shoulder width is a vendor option, either half or full width.



4220202/B 12/2023

NOTES:

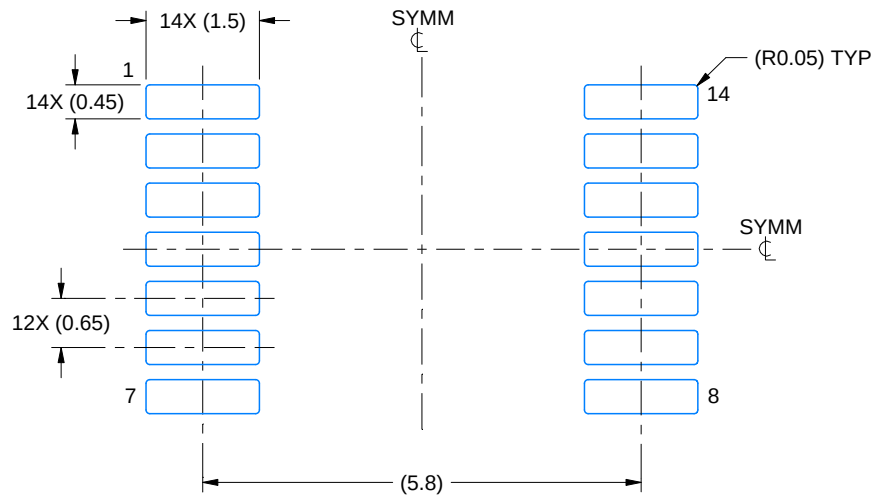
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

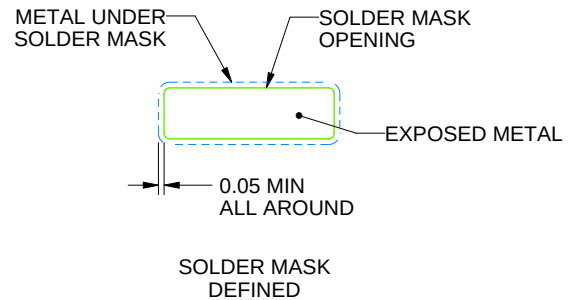
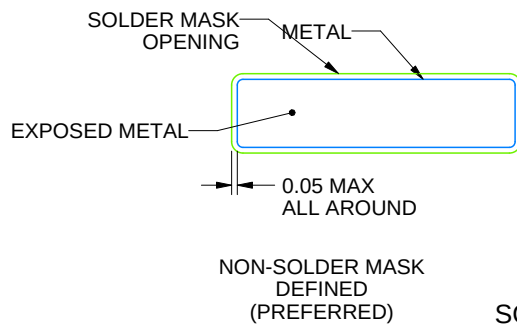
PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X



SOLDER MASK DETAILS

4220202/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

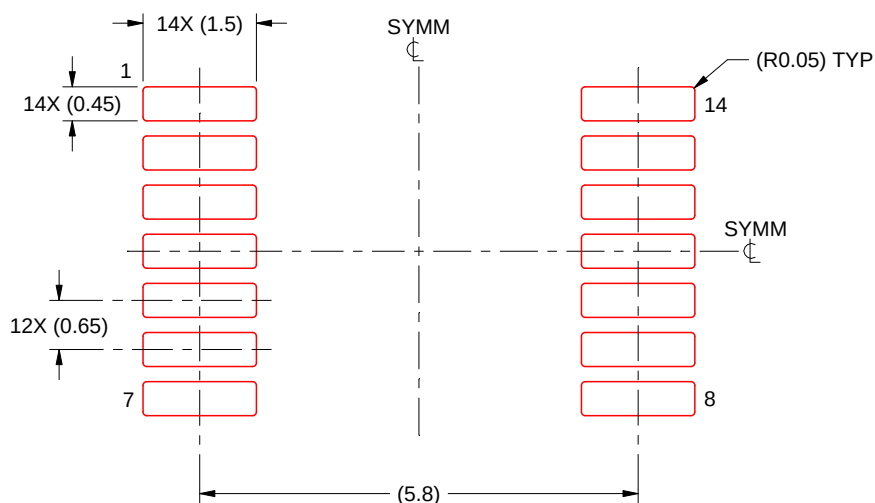
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220202/B 12/2023

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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Last updated 10/2025